

Computational Social Science with Images and Audio

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08 October 2026

What We Will Do

1. Understand how CLIP learns a shared representation of images and language
2. Use PCA and k -means to explore structure in embedding spaces
3. See why a useful CLIP vector remains hard to interpret
4. Derive SpLiCE as a sparse, nonnegative recovery problem
5. Use concept weights to audit data and intervene on representations

The class moves from prediction to interpretation, then back to research design.

Outline

- 1 CLIP
- 2 PCA and Clustering
- 3 SpLiCE
- 4 Applications
- 5 Takeaways

Why CLIP Is Different

A traditional image classifier takes image I_i and predicts class Y_i from K labels fixed during training:

$$I_i \longrightarrow \Pr(Y_i \in \{1, \dots, K\} \mid I_i).$$

CLIP instead compares image I_i with researcher-supplied text T_j :

$$(I_i, T_j) \longrightarrow \text{sim}(I_i, T_j).$$

- After pretraining, researchers can supply new descriptions without retraining CLIP
- Similarity scores can rank images, assign labels, or become variables in an analysis
- The result is conditional on the descriptions: changing the wording can change the measurement

CLIP: Images and Text in a Shared Space

CLIP trains an image encoder f_I and a text encoder f_T :

$$u_i = \frac{f_I(I_i)}{\|f_I(I_i)\|_2}, \quad v_i = \frac{f_T(T_i)}{\|f_T(T_i)\|_2}, \quad u_i, v_i \in \mathbb{R}^d.$$

Their similarity is a temperature-scaled dot product:

$$s_{ij} = \frac{u_i^\top v_j}{\tau}.$$

- u_i and v_i are unit embeddings in d dimensions
- Normalization makes their dot product equal cosine similarity
- $\tau > 0$ is the temperature: smaller values sharpen similarity differences
- Matched image–text pairs should have high similarity

How Is CLIP Trained?

For N matched pairs, i indexes the query, j a candidate match, and m all candidates in the denominator:

$$p_{ij}^{I \rightarrow T} = \frac{e^{s_{ij}}}{\sum_{m=1}^N e^{s_{im}}}, \quad p_{ij}^{T \rightarrow I} = \frac{e^{s_{ij}}}{\sum_{m=1}^N e^{s_{mj}}}.$$

The loss rewards the correct pair in both directions:

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{2N} \sum_{i=1}^N [\log p_{ii}^{I \rightarrow T} + \log p_{ii}^{T \rightarrow I}].$$

- The numerator is the observed match; the denominator compares it with all N candidates
- $p_{ii}^{I \rightarrow T}$ asks image i to choose caption i ; $p_{ii}^{T \rightarrow I}$ reverses the task
- Minimizing $-\log p$ pushes the matched probability toward one; smaller τ sharpens the competition

Zero-Shot Classification

Given class names $y \in \mathcal{Y}$, write one or more prompts for every class:

$$v_y = f_T(\text{"a photo of a } y\text{"}), \quad \hat{y}_i = \arg \max_{y \in \mathcal{Y}} u_i^\top v_y.$$

$\mathcal{Y}$ is the candidate label set, v_y is the prompt embedding for class y , and $\hat{y}_i$ is the predicted label for image i .



- “border security”
- “migration”
- “military operations”

The ranking is conditional on this candidate set and the prompt wording.

Prompt Choice Is a Measurement Choice

The same intended construct can be expressed in different ways:

“migration” (bare label)

“a photo about migration” (generic template)

“a television news image about migration” (domain template)

- Prompt ensembles reduce sensitivity to a single wording
- Domain-specific prompts can improve construct validity
- Human labels remain necessary for validation

Useful Predictions, Opaque Coordinates

CLIP returns a dense vector:

$$z^{\text{img}} = (0.2435, 0.5642, -0.0183, \dots, 0.4633) \in \mathbb{R}^d.$$

Here d is the embedding dimension.

- Nearby vectors often share semantic content
- Individual coordinates do not have stable human-readable meanings
- A prediction can therefore be accurate without explaining which concepts drive it

Before SpLiCE, unsupervised methods can reveal structure in the embedding geometry.

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Unsupervised Learning: Finding Structure in X

Stack N observations as rows of an embedding matrix with p features:

$$X = (x_1, \dots, x_N)' \in \mathbb{R}^{N \times p}, \quad x_i \in \mathbb{R}^p \text{ is observation } i.$$

- Unsupervised learning uses X without an observed outcome Y
- **PCA** replaces p variables with a small number of continuous components
- **k -means** replaces N observations with membership in K researcher-chosen groups
- Both methods summarize the geometry of a representation before substantive interpretation

PCA summarizes directions of variation. k -means summarizes groups of observations.

Unsupervised Tasks: PCA

- PCA finds a lower-dimensional representation that retains as much of the variance as possible
- Approximate each observation with $x_i \approx \mu + V_q \lambda_i$, where V_q is a $p \times q$ matrix mapping q component scores into the original p -dimensional space
- Solve $\min_{\lambda, V_q, \mu} \sum_{i=1}^N \|x_i - \mu - V_q \lambda_i\|^2$ subject to $V_q' V_q = I_q$
- Compute the singular value decomposition of the centered data matrix: $X_c = UDV^T$
- The columns of V are the eigenvectors of $X_c^T X_c$ and give directions in the original feature space, ordered by explained variance
- The first component captures the most variance; the first q components retain the greatest possible variance among linear q -dimensional projections

Unsupervised Tasks: k -Means Clustering

Choose K groups. C_j is the set of observations assigned to cluster j , and $\mu_j \in \mathbb{R}^p$ is its center:

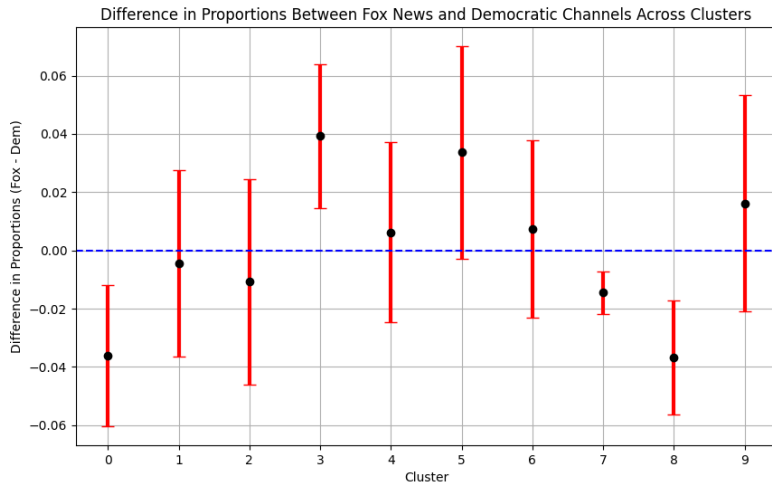
$$\min_{C_1, \dots, C_K, \mu_1, \dots, \mu_K} \sum_{j=1}^K \sum_{i \in C_j} \|x_i - \mu_j\|_2^2.$$

$|C_j|$ is the number of observations assigned to cluster j . For fixed assignments, the optimal center is the within-cluster mean:

$$\mu_j = \frac{1}{|C_j|} \sum_{i \in C_j} x_i.$$

- Assignment step: send each observation to its nearest center
- Update step: recompute every center from its assigned observations
- Iterate until assignments stop changing or the objective stabilizes
- Interpret a cluster by inspecting observations nearest its center

Application: Comparing Image Clusters Across Channels



What Does a Cluster Contain?

Images from Cluster 3 - Fox News



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SpLiCE

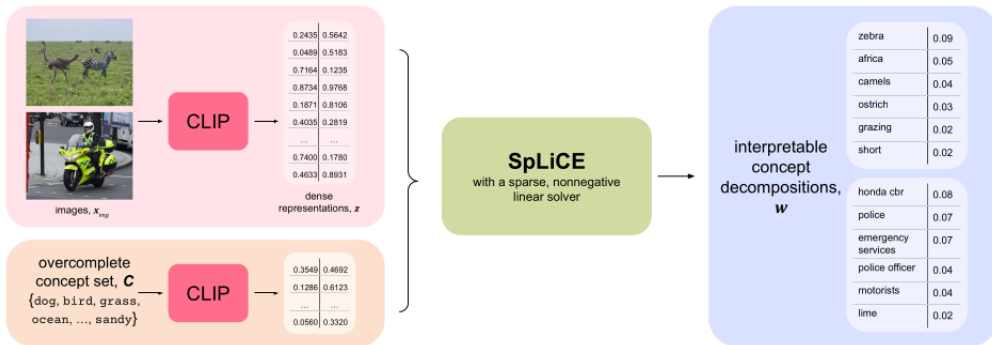
Bhalla et al. (2024), NeurIPS

Question

Can a dense CLIP representation be decomposed into a small set of human-readable semantic concepts?

- Post hoc: CLIP remains fixed.
- Task agnostic: the decomposition does not depend on a downstream classifier.
- Output: a short list of concepts and their weights.

SpLiCE in One Picture



Dense CLIP vector + text concept dictionary $\rightarrow$ sparse nonnegative weights.

Notation

Object	Definition	Role
$z^{\text{img}} \in \mathbb{R}^d$	CLIP image embedding	Original dense representation
$z \in \mathbb{R}^d$	Centered, normalized image embedding	Target of sparse recovery
$C \in \mathbb{R}^{d \times c}$	Centered concept dictionary	One text concept per column
$w \in \mathbb{R}_+^c$	Nonnegative concept weights	Sparse explanation
$\hat{z}^{\text{img}}$	Reconstructed image embedding	Returned to the original CLIP space

The central approximation is $z \approx Cw$ with $\|w\|_0 \ll c$. Here d is the embedding dimension, c the dictionary size, and $\|w\|_0$ counts nonzero weights.

A Latent Concept Data-Generating Process

Let $\omega \in \mathbb{R}_+^k$ encode semantic concepts and $\epsilon \in \mathbb{R}^\ell$ encode nuisance variation:

$$x^{\text{img}} = h^{\text{img}}(\omega, \epsilon), \quad x^{\text{txt}} = h^{\text{txt}}(\omega, \epsilon), \quad \omega \sim \rho, \quad \epsilon \sim \phi.$$

The two CLIP encoders are

$$z^{\text{img}} = f(x^{\text{img}}), \quad z^{\text{txt}} = g(x^{\text{txt}}).$$

- k is the number of possible concepts; ℓ is the nuisance dimension; $\omega_j \geq 0$ is the strength of concept j
- ρ and ϕ are the distributions of semantic and nuisance variables
- $h^{\text{img}}, h^{\text{txt}}$ generate pixels and words; f, g map them to CLIP embeddings

Goal: recover $f(x^{\text{img}})$ using text concept directions.

Sufficient Conditions I: Sparsity and Invariance

Assumption 1: sparse concept support

$$\|\omega\|_0 \leq \alpha, \quad \alpha \ll k.$$

$\|\omega\|_0$ counts active concepts, α is their maximum number, and k is the total number available.

Intuition: One image should contain only a few concepts from the full vocabulary.

Assumption 2: invariance to nuisance variation

$$f(h^{\text{img}}(\omega, \epsilon)) = f(h^{\text{img}}(\omega, \epsilon')) \quad \forall \epsilon, \epsilon',$$

ϵ and ϵ' change nuisance details while ω stays fixed. The same condition applies to text.

Intuition: Changing lighting, viewpoint, or wording should not change the embedding when meaning stays fixed.

Sufficient Conditions II: Linearity and Alignment

Assumption 3: linearity in concept space

$$f \circ h^{\text{img}} \quad \text{and} \quad g \circ h^{\text{txt}} \quad \text{are linear in } \omega.$$

$f \circ h^{\text{img}}$ means generate the image with h^{img} , then encode it with f .

Intuition: A scene with “snow” and “mountain” should combine those two concept directions.

Assumption 4: aligned image and text encoders

$$f(h^{\text{img}}(\omega, \epsilon)) = g(h^{\text{txt}}(\omega, \epsilon)) \quad \text{for a given } \omega.$$

Intuition: An image and a caption expressing the same concepts should map to the same point. Raw CLIP only approximates this condition. SpLiCE centers each modality to reduce the gap.

What the Assumptions Imply

By invariance, suppress ϵ . Let $e_j \in \mathbb{R}^k$ be the vector with a one in position j and zeros elsewhere:

$$\omega = \sum_{j=1}^k \omega_j e_j.$$

Linearity gives

$$\begin{aligned} f(x^{\text{img}}) &= f(h^{\text{img}}(\omega)) \\ &= \sum_{j=1}^k \omega_j f(h^{\text{img}}(e_j)) \\ &= C^{\text{img}} \omega. \end{aligned}$$

Alignment gives $C^{\text{img}} = C^{\text{txt}}$, hence

$$f(x^{\text{img}}) = C^{\text{txt}} \omega, \quad \|\omega\|_0 \leq \alpha.$$

$C^{\text{img}} = [f(h^{\text{img}}(e_1)), \dots, f(h^{\text{img}}(e_k))]$; C^{txt} is defined analogously.

A Sanity Check for Approximate Linearity

Let z_a, z_b be embeddings of separate inputs and z_{ab} the embedding of their combination. Fit weights w_a, w_b so that

$$w_a z_a + w_b z_b \approx z_{ab}.$$

The fitted reconstruction is $\hat{z}_{ab} = w_a z_a + w_b z_b$; $\cos(\hat{z}_{ab}, z_{ab})$ measures directional similarity.

	w_a	w_b	$\cos(\hat{z}_{ab}, z_{ab})$
ImageNet	0.48 ± 0.09	0.45 ± 0.09	0.76 ± 0.05
CIFAR100	0.45 ± 0.08	0.42 ± 0.08	0.75 ± 0.03
MIT States	0.48 ± 0.09	0.45 ± 0.09	0.76 ± 0.05
COCO Text	0.59 ± 0.12	0.47 ± 0.12	0.88 ± 0.04

The result supports approximate composition. It does not establish exact linearity. Bhalla et al. (2024), Table 1

The Concept Dictionary

The vocabulary $\mathcal{V}$ contains c phrases; column C_j is the centered text embedding of phrase j :

$$\mathcal{V} = \{\text{dog, tabby cat, cloudy, birthday party, } \dots\}.$$

- Start from frequent one- and two-word phrases in LAION-400M captions
- Remove NSFW samples and near-duplicate concepts
- Prune pairs with cosine similarity above 0.9
- Keep 10,000 one-word concepts and 5,000 two-word concepts

The dictionary defines which meanings the explanation can express.

Why Sparse and Nonnegative?

Sparsity

$$\|w\|_0 = \#\{j : w_j \neq 0\} \ll c.$$

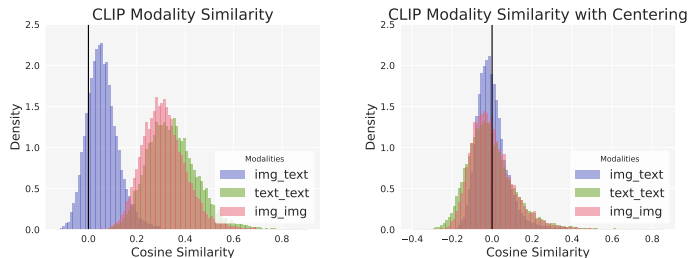
- Most concept weights equal zero
- A short list is easier to inspect
- The paper targets roughly 10–30 active concepts

Nonnegativity

$$w \in \mathbb{R}_+^c \iff w_j \geq 0 \ \forall j.$$

- Present concepts contribute positively
- “Not tiger” need not equal negative “tiger”
- Antonyms can enter as separate dictionary atoms

CLIP Still Has a Modality Gap



- Image–image and text–text similarities concentrate near 0.3
- Image–text similarities concentrate closer to zero
- The modalities occupy different cones on the unit sphere

Centering and Renormalizing

Define $\sigma(a) = a / \|a\|_2$. The centered image target is

$$z = \sigma(z^{\text{img}} - \mu_{\text{img}}).$$

For concept phrase x_j^{con} , dictionary column j is

$$C_j = \sigma(g(x_j^{\text{con}}) - \mu_{\text{con}}), \quad C = [C_1, \dots, C_c].$$

- $j = 1, \dots, c$ indexes concepts and g is CLIP's text encoder
- μ_{img} is estimated from MSCOCO images
- μ_{con} is the mean of the concept vocabulary embeddings

The Ideal Sparse Recovery Problem

The target formulation asks for the smallest concept set that preserves direction:

$$\min_{w \in \mathbb{R}_+^c} \|w\|_0 \quad \text{subject to} \quad \left\langle z, \frac{Cw}{\|Cw\|_2} \right\rangle \geq 1 - \varepsilon.$$

- $\|w\|_0$ counts active concepts
- ε sets the tolerated loss in cosine similarity
- $\langle a, b \rangle = a^\top b$ is the dot product; here it is cosine similarity
- Direct ℓ_0 optimization is combinatorial

From the ℓ_0 Constraint to an ℓ_1 Penalty

Replace the hard sparsity problem with a penalized cosine objective:

$$\max_{w \in \mathbb{R}_+^c} \left\langle z, \frac{Cw}{\|Cw\|_2} \right\rangle - \lambda \|w\|_1.$$

Here $\lambda \geq 0$ controls sparsity and $\|w\|_1 = \sum_j |w_j| = \sum_j w_j$ because $w \geq 0$. For any unit vectors x, y ,

$$\|x - y\|_2^2 = 2 - 2\langle x, y \rangle.$$

Therefore the same directional fit can be written as

$$\min_{w \in \mathbb{R}_+^c} \left\| \frac{Cw}{\|Cw\|_2} - z \right\|_2^2 + 2\lambda \|w\|_1.$$

The Convex Relaxation Used by SpLiCE

With C and z fixed, the normalized problem remains nonconvex. SpLiCE fits

$$w^* = \arg \min_{w \in \mathbb{R}_+^c} \|Cw - z\|_2^2 + 2\lambda \|w\|_1.$$

Reconstruction term

$$\|Cw - z\|_2^2$$

fits the centered image vector.

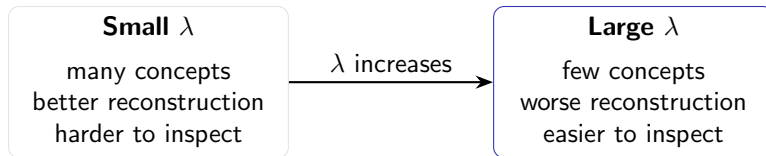
This is a nonnegative LASSO problem.

Sparsity term

$$2\lambda \|w\|_1$$

uses λ to shrink unnecessary weights to zero.

What Does λ Control?



- Report results against the realized ℓ_0 norm, the number of active weights
- Choose a readable range and verify downstream performance
- Reconstruction quality alone does not determine the best explanation

Returning to the Original CLIP Space

Optimization produces w^* in the centered space:

$$\tilde{z} = Cw^*.$$

SpLiCE adds the image mean back and normalizes:

$$\hat{z}^{\text{img}} = \sigma(Cw^* + \mu_{\text{img}}).$$

Here $\sigma(a) = a / \|a\|_2$ and $\tilde{z}$ denotes the centered reconstruction.

- w^* is the readable explanation
- Cw^* is the centered dense reconstruction
- $\hat{z}^{\text{img}}$ can replace the original embedding in downstream CLIP tasks

Reading a SpLiCE Decomposition



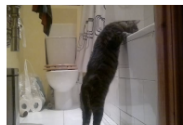
"A statue of a cow sitting under a sign on grass."

wedding sign	0.06
cow	0.04
shop logo	0.04
sweets	0.04
sweetened	0.03
swing	0.02



"A field that has some snow and a man skiing in it"

powder	0.08
snowboard	0.07
skiing	0.07
slopes	0.06
winter sports	0.04
ski resort	0.03



"A cat is looking at a bathtub in a bathroom with a toilet."

cat	0.05
shower	0.05
tabby cat	0.03
scratching	0.03
shampoo	0.03
sink	0.02



"A knife stabbing an orange with a sad face."

vitamin c	0.10
oranges	0.08
fresh orange	0.06
depression	0.04
unhappy	0.04
food background	0.03
macro shot	0.03



"Adult with orange umbrella walks on bricked open area."

red umbrella	0.07
umbrella	0.06
woman walks	0.03
businessman	0.03
autumn winter	0.01
orange	0.01
background	0.01
paved	0.01

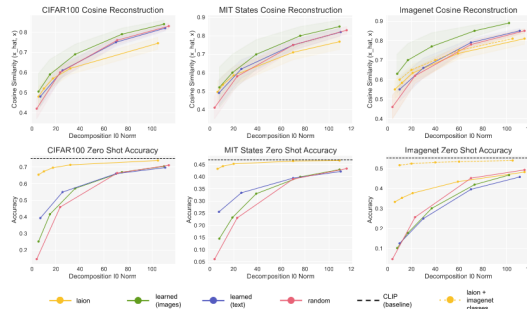


"A foal on the beach laying next to a horse."

wild horses	0.10
blackandwhite	0.08
bw	0.07
mare	0.07
beach sand	0.03
horse	0.02
resting	0.01
kissing	0.01

Figure 2: Example images from MSCOCO shown with their captions below and their concept decompositions on the right. We display the top seven concepts for visualization purposes, but images in the figure had decompositions with 7-20 concepts.

Interpretability and Performance Trade Off

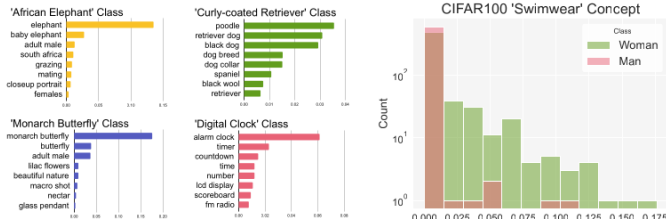


- Top row: cosine reconstruction of the original CLIP vector
- Bottom row: zero-shot classification accuracy
- The LAION dictionary preserves much of the semantic performance before exact reconstruction

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Auditing Datasets for Spurious Correlations



- Aggregate concept weights by class, then compare related classes
- SpLiCE surfaced “swimwear” and “bra” for the CIFAR100 woman class
- Manual inspection found at least 70 of 600 images with bikinis, underclothes, or partial undress

Intervening on the “Glasses” Concept

1. Classify the original representation

$$\hat{y}_a = \arg \max_{k \in \{0,1\}} \langle \hat{z}^{\text{img}}, t_{a,k} \rangle.$$

a is the task (glasses or gender), k is a possible label, and $t_{a,k}$ is its CLIP text embedding. Accuracy uses the CelebA labels.

Task	Before	After
Glasses	0.88	0.69
Gender	0.97	0.96

2. Delete only the glasses coordinate

$$w_{\text{glasses}}^* \leftarrow 0, \quad \hat{z}_{-\text{glasses}}^{\text{img}} = \sigma(Cw_{-\text{glasses}}^* + \mu_{\text{img}}).$$

Removing one concept sharply reduces glasses accuracy while leaving gender accuracy almost unchanged.

3. Classify again

Use the same text labels and CelebA targets. Only the image representation has changed.

Bhalla et al. (2024), Table 2

Limits of a SpLiCE Audit

- **Dictionary:** LAION captions carry omissions, stereotypes, and language-specific priors
- **Concept length:** one- and two-word atoms miss longer and relational meanings
- **Specification:** results depend on centering data, vocabulary pruning, and λ
- **Fidelity:** a readable reconstruction can omit dense information used downstream
- **Inference:** concept deletion does not automatically identify a causal effect outside the model

Interpretability changes the questions we can ask. Validation remains necessary.

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A Social Science Workflow

1. **Construct:** define the theoretical concept and unit of analysis
2. **Sample:** document how images entered the corpus
3. **Specify:** preregister prompts, concept vocabularies, and aggregation rules
4. **Validate:** compare with human labels and subgroup-specific errors
5. **Infer:** carry measurement uncertainty into the final design

CLIP and SpLiCE enter at step 3. Credibility depends heavily on the other four steps.

What to Remember

- **CLIP:** language becomes an interface for comparing and measuring visual content
- **SpLiCE:** a centered text dictionary turns some dense geometry into sparse concept weights
- **Mathematics:** sparsity, approximate linearity, and cross-modal alignment justify the decomposition
- **Research design:** prompts and dictionaries are coding choices that require validation

What would you trust CLIP to measure in your own research?

Appendix: How the Optimization Is Solved

The paper implements batched ADMM for the nonnegative LASSO objective:

$$\min_{w \in \mathbb{R}_+^c} \|Cw - z\|_2^2 + 2\lambda \|w\|_1.$$

ADMM separates the smooth quadratic term from sparsity and nonnegativity:

1. Solve a quadratic update for the reconstruction weights
2. Apply soft thresholding and project onto the nonnegative orthant
3. Update the dual variable and repeat until primal and dual residuals are small

The implementation uses $\rho_{\text{ADMM}} = 5$, tolerance 10^{-4} , and reports convergence within 1,000 iterations for batches of 1,024. Here ρ_{ADMM} is the algorithm's penalty parameter, not the latent distribution ρ used earlier. Bhalla et al. (2024), Appendix A.3

Appendix: The Relaxed Objective as Shrinkage

Expand the squared error:

$$\|Cw - z\|_2^2 + 2\lambda \|w\|_1 = \|Cw\|_2^2 + \|z\|_2^2 - 2\langle Cw, z \rangle + 2\lambda \|w\|_1.$$

Since $\|z\|_2 = 1$, minimizing is equivalent to maximizing

$$\langle Cw, z \rangle - \frac{1}{2} \|Cw\|_2^2 - \lambda \|w\|_1.$$

- The first term rewards alignment with the image
- The second shrinks the reconstruction norm
- The third rewards sparse concept weights

The method normalizes after solving because Cw is generally low norm. Bhalla et al. (2024), Appendix A.2